MODEL SELECTION NOTE

BSE4104 EMERGING TRENDS IN SOFTWARE ENGINEERING

University Student-Support Case Agent

BY GROUP M EVENING

Model Chosen: Google Gemini Flash (via the Gemini Developer API / Google AI Studio)

Capability: Gemini Flash offers strong general-purpose language understanding and instruction-following, sufficient for our core tasks: interpreting freshman onboarding questions, retrieving and synthesizing grounded answers, and reasoning about when to escalate to a support case. It supports a large context window (up to 1M tokens), giving comfortable headroom for our ~10–50 document RAG corpus as the project develops.

Cost: Gemini Flash provides a free-tier option suitable for our project without requiring us to commit to paid API usage. This matters given our budget constraints as a student team, and the volume of repeated testing this project requires.

Latency: Flash-tier models are optimized for speed over maximum reasoning depth, which suits our use case (US-1) which commits to answering onboarding questions within 10 seconds, and Flash comfortably supports that.

Privacy: Since our knowledge base is built entirely from public university documents and our case data is team-created synthetic data (no real student data), the standard free-tier trade-off that Google may use prompt/output data to improve their models is an acceptable risk for this project. No confidential or personally identifiable information will pass through the API at any point.

Access: Setup is straightforward: team members obtain an API key directly through Google AI Studio using a Google account, with no card, billing setup, or application process required.

Alternatives Considered

- **Claude Haiku 4.5**: stronger instruction-following, but no free API tier; requires a paid account
- **GPT-4o-mini**: competitively priced, but no current free trial credits; also requires a card on file
- **OpenRouter (free models)**: useful for experimenting with multiple models through a single API, but free-model availability and request limits can constrain sustained development and evaluation.

Gemini Flash was selected because it provides a strong balance of capability, latency, accessibility, and cost for an eight-week student project. It is sufficiently capable for our RAG and support-case workflow while avoiding a dependency on paid API usage during development and evaluation.